

DRY-RUN GUIDE | 06_largest_number.cpp
Source: cpp\2nd_belt\06_largest_number.cpp

CORE BEHAVIOR

Reads the first value as the initial maximum, then scans remaining values and replaces the maximum only when a larger value appears.

REPRESENTATIVE INPUT / CALL

Input: N=5; values '-4 7 2 12 3'.

STEP TRACE

Start largest=-4; 7 replaces it; 2 does not; 12 replaces it; 3 does not.

EXPECTED OUTPUT / RETURN

'12'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | Given a list of N integers, find the largest number in the list.
004 | Print the largest element.
005 |
006 | Input:
007 | The first line contains an integer N representing the number of elements.
008 | The second line contains N space-separated integers.
009 |
010 | Output:
011 | Print the largest number in the list.
012 | */
013 |
014 | #include <iostream>
015 | using namespace std;
016 |
017 | int main() {
018 |     int n;
019 |     cin >> n;
020 |
021 |     int x;
022 |     cin >> x;
023 |     int largest = x;
024 |
025 |     for (int i = 1; i < n; i++) {
026 |         cin >> x;
027 |         if (x > largest) {
028 |             largest = x;
029 |         }
030 |     }
031 |
032 |     cout << largest;
033 |
034 |     return 0;
035 | }
036 |
```